IILM UNIVERSITY

Greater Noida, Uttar Pradesh

INTERNSHIP EVALUATION PRESENTATION

Artificial Neural Network and its Application

STUDENT

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ENROLLMENT NO.

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PROGRAM & SEMESTER

B.Tech CSE, 5th Semester

ORGANIZATION

Banaras Hindu University, Varanasi

INTERNSHIP SNAPSHOT

Internship Overview

1 Host Institution

Dept. of Computer Science, Institute of Science, Banaras Hindu University (BHU), Varanasi

2 Duration

8 Weeks
June 1 – July 25, 2026

3 Supervisor

Prof. Manoj Kumar Singh
Dept. of Computer Science, BHU

4 Nature

Research Internship — Study & implementation of Artificial Neural Networks

Problem Statement & Objectives

PROBLEM STATEMENT

Conventional rule-based programs struggle with tasks that involve pattern recognition, non-linear relationships, or noisy real-world data — such as classification and prediction problems. The internship explored how Artificial Neural Networks (ANNs), inspired by the human brain, can learn such patterns directly from data.

OBJECTIVES

1. Understand the theoretical foundation of ANNs and biological inspiration
2. Study the architecture: neurons, layers, weights & biases
3. Implement a feed-forward neural network from scratch in Python
4. Train the model using forward propagation & backpropagation
5. Evaluate performance and explore real-world applications

THE FOUNDATION

What is an Artificial Neural Network?

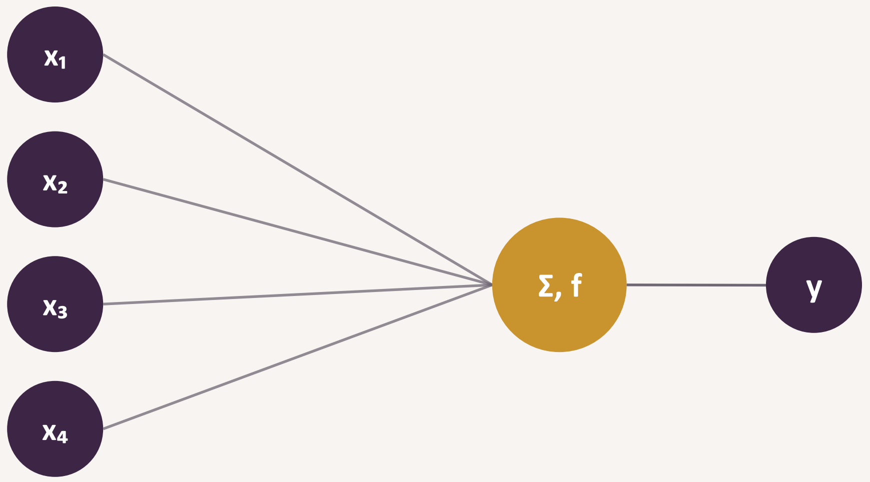
An Artificial Neural Network (ANN) is a computing system loosely inspired by the network of neurons in the human brain. It consists of interconnected processing units (“neurons”) arranged in layers, which learn to map inputs to outputs by adjusting the strength of connections between them — without being explicitly programmed with rules.

✓ **Learns from data**
Improves performance through repeated exposure to examples

✓ **Handles non-linearity**
Captures complex relationships simple equations cannot

✓ **Generalizes**
Makes reasonable predictions on unseen inputs

A SINGLE ARTIFICIAL NEURON



Weighted inputs are summed, passed through an activation function f, and produce an output signal y.

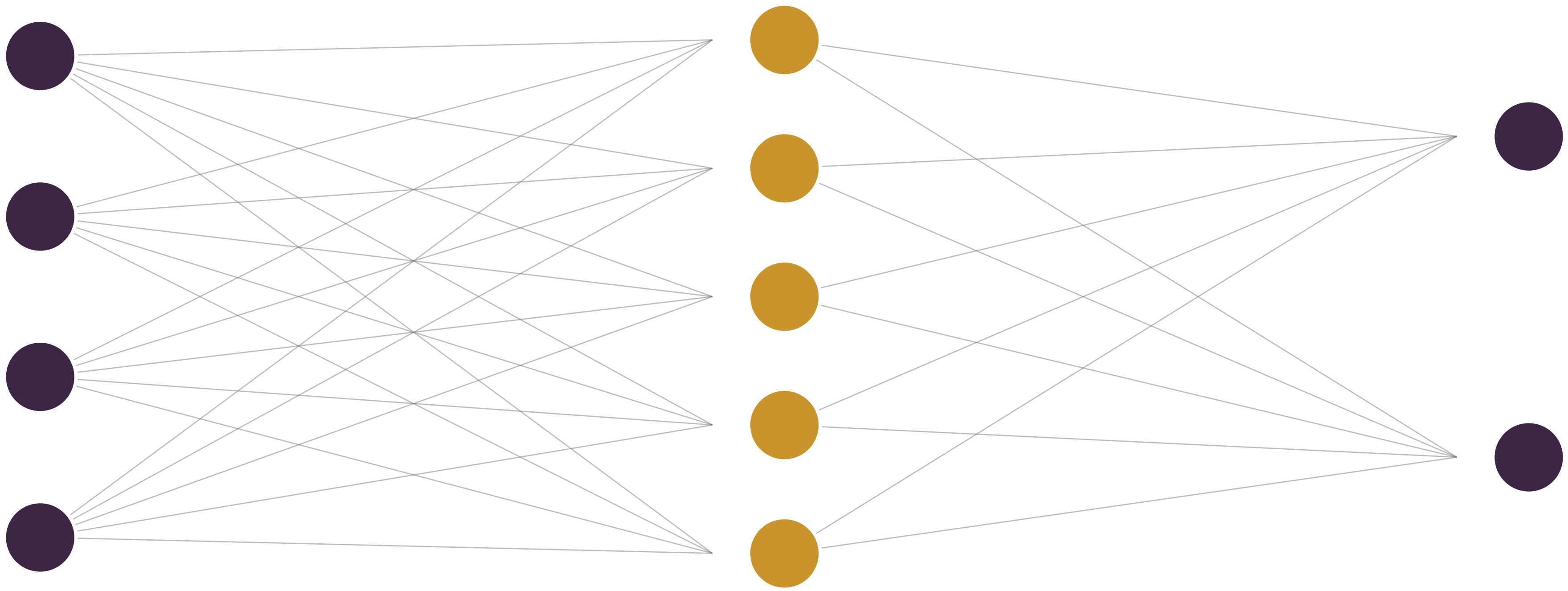
INSPIRATION

Biological Neuron vs. Artificial Neuron

Biological Neuron		Artificial Neuron
Dendrites Receive signals from other neurons	⟷	**Inputs (x)** Numerical values fed into the neuron
Cell Body (Soma) Aggregates incoming electrical signals	⟷	**Summation (Σ w·x + b)** Weighted sum of inputs plus bias
Axon Transmits signal if a threshold is crossed	⟷	**Activation Function** Decides the neuron's output signal
Synapse Connection strength between neurons	⟷	**Weight** Learnable parameter — analogous to synapse strength

ARCHITECTURE

Structure of a Feed-Forward Neural Network



Input Layer

Features (x_1…x_4)

Hidden Layer

Learns internal patterns

Output Layer

Predicted class / value

NON-LINEARITY

Activation Functions Studied

1

Sigmoid

$f(x) = 1 / (1 + e^{-x})$

Squashes output to (0, 1); used for binary probabilities

2

ReLU

$f(x) = \max(0, x)$

Fast, avoids vanishing gradients; default for hidden layers

3

Tanh

$f(x) = \tanh(x)$

Zero-centred output in (-1, 1); often faster convergence

HOW THE NETWORK LEARNS

Forward Propagation & Backpropagation

FORWARD PROPAGATION

Input Data → Weighted Sum → Activation → Prediction (ŷ)

BACKPROPAGATION • GRADIENT DESCENT

Compute Loss → Error Gradient → Update Weights → Repeat (Epochs)

The network makes a prediction (forward pass), measures how wrong it is using a loss function, then propagates that error backward to update every weight in the direction that reduces the loss (gradient descent). Repeating this over many epochs allows the network to gradually improve its predictions.

CHAPTER 4.6

Tools & Technologies Used

Py

Python

Core programming language for implementation

Np

NumPy

Matrix operations, weight initialization & computation

Pd

Pandas

Loading, cleaning and structuring the dataset

Mp

Matplotlib

Visualizing training loss and accuracy curves

Jn

Jupyter Notebook

Interactive development and experimentation

Vs

VS Code

Script writing, debugging and version control

Methodology / Work Flow

1. **Data Collection** — Gathered a labelled dataset suited for a classification task
2. **Preprocessing** — Cleaned data, handled missing values, normalized features with Pandas/NumPy
3. **Model Design** — Defined a feed-forward ANN — input, hidden and output layers with chosen activation functions
4. **Training** — Ran forward + backward propagation over multiple epochs to minimize loss
5. **Evaluation** — Measured accuracy / loss and visualized training curves with Matplotlib

REAL-WORLD IMPACT

Applications of Artificial Neural Networks

1

Image & Pattern Recognition

Facial emotion detection, object and handwriting recognition

2

Predictive Systems

Forecasting continuous values such as air quality index (AQI) or demand

3

Natural Language Processing

Chatbots and conversational assistants that understand user queries

4

Healthcare

Assisting diagnosis and medical guidance systems

5

Finance

Fraud detection and stock trend prediction

6

Autonomous Systems

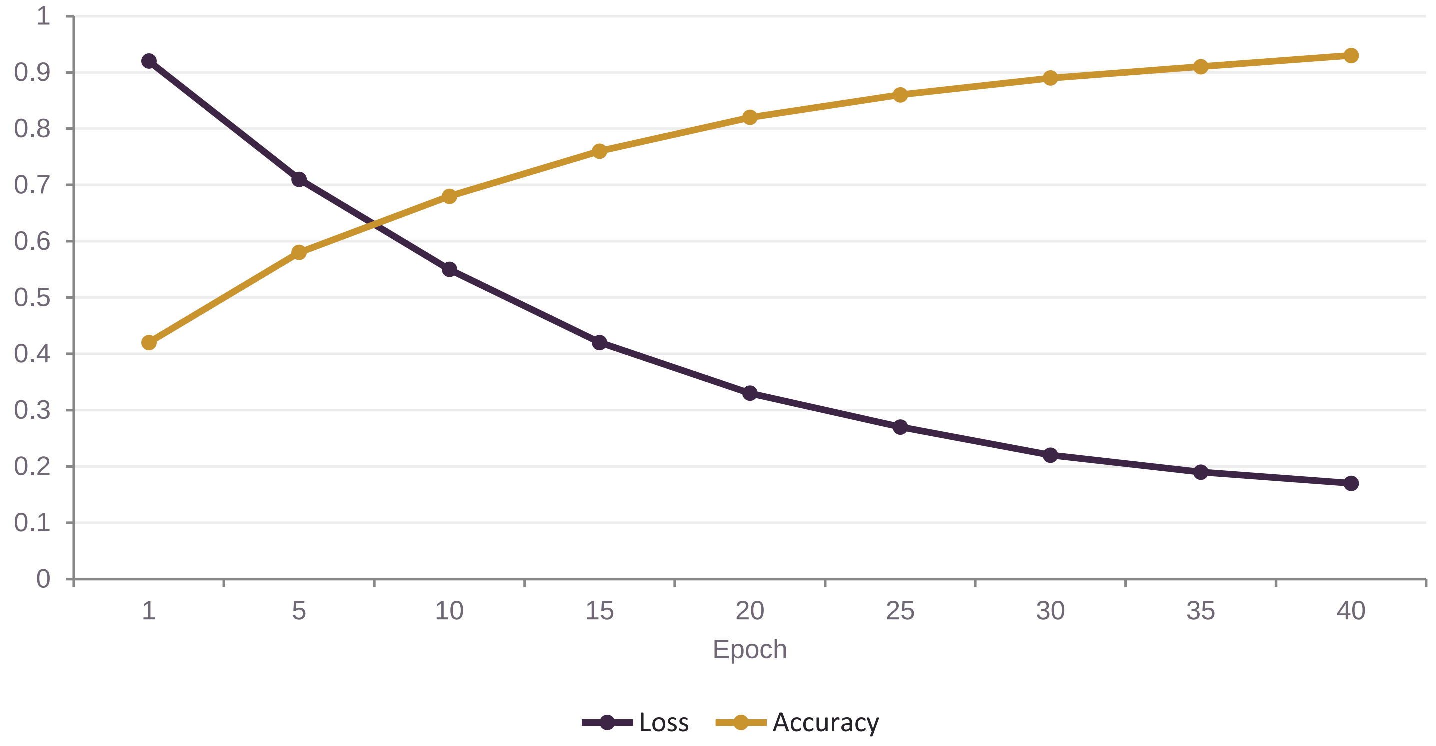
Perception and decision-making in self-driving vehicles & robotics

CHAPTER 4.9

Results & Observations

The implemented network showed a steady decrease in training loss and a corresponding rise in accuracy across epochs, confirming that the model was successfully learning patterns from the data.

TRAINING PROGRESS (ILLUSTRATIVE)



- Loss decreased consistently, indicating correct backpropagation implementation
- Accuracy improved with more training epochs, then plateaued
- Choice of activation function affected convergence speed
- Learning rate had to be tuned to avoid slow or unstable training

ROADBLOCKS

Challenges Faced

! Mathematical Foundation

Understanding matrix calculus behind backpropagation required extra study of linear algebra and derivatives.

! Debugging From Scratch

Implementing the network without a high-level framework meant every bug (shape mismatches, wrong gradients) had to be traced manually.

! Hyperparameter Tuning

Selecting an appropriate learning rate, number of hidden neurons and epochs took several rounds of experimentation.

! Overfitting

The model sometimes performed well on training data but poorly on unseen data, requiring regularization awareness.

TAKEAWAYS

Key Learnings & Skills Gained

- ✓ Solid conceptual understanding of how neural networks learn
- ✓ Hands-on experience implementing ML concepts in Python & NumPy
- ✓ Improved data-handling skills using Pandas
- ✓ Ability to visualize and interpret model performance with Matplotlib
- ✓ Exposure to academic research methodology under expert guidance
- ✓ Stronger foundation for advanced AI/ML coursework and projects

CONCLUSION

This internship provided a strong foundation in Artificial Neural Networks — from biological inspiration to hands-on implementation of forward propagation and backpropagation. Beyond the technical skills, it strengthened problem-solving ability and research thinking, both of which will support future work in Artificial Intelligence and Machine Learning.

8	4	3+
Weeks of Research	ANN Concepts Mastered	Real-world Applications Explored

Thank You

Questions & Discussion